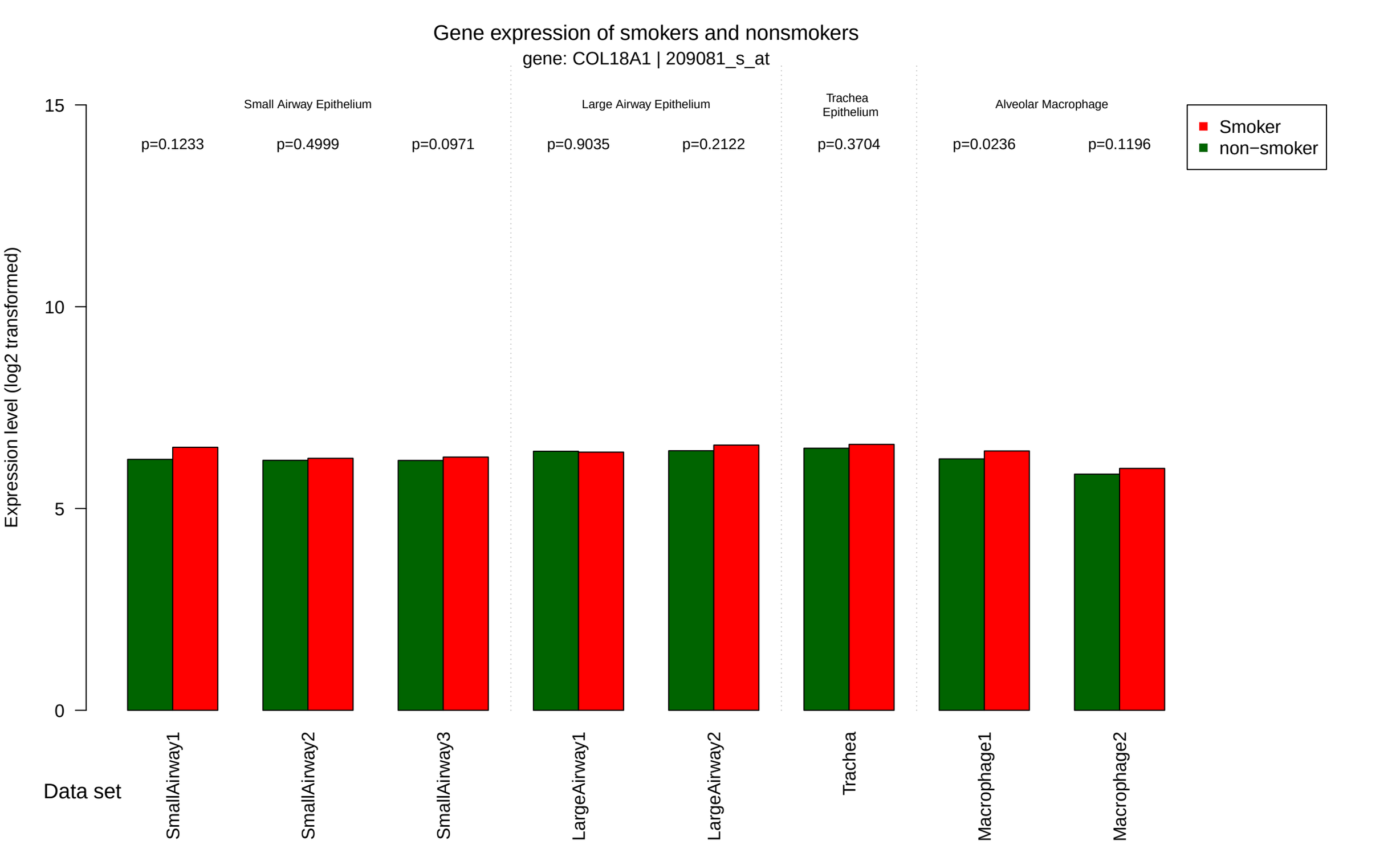
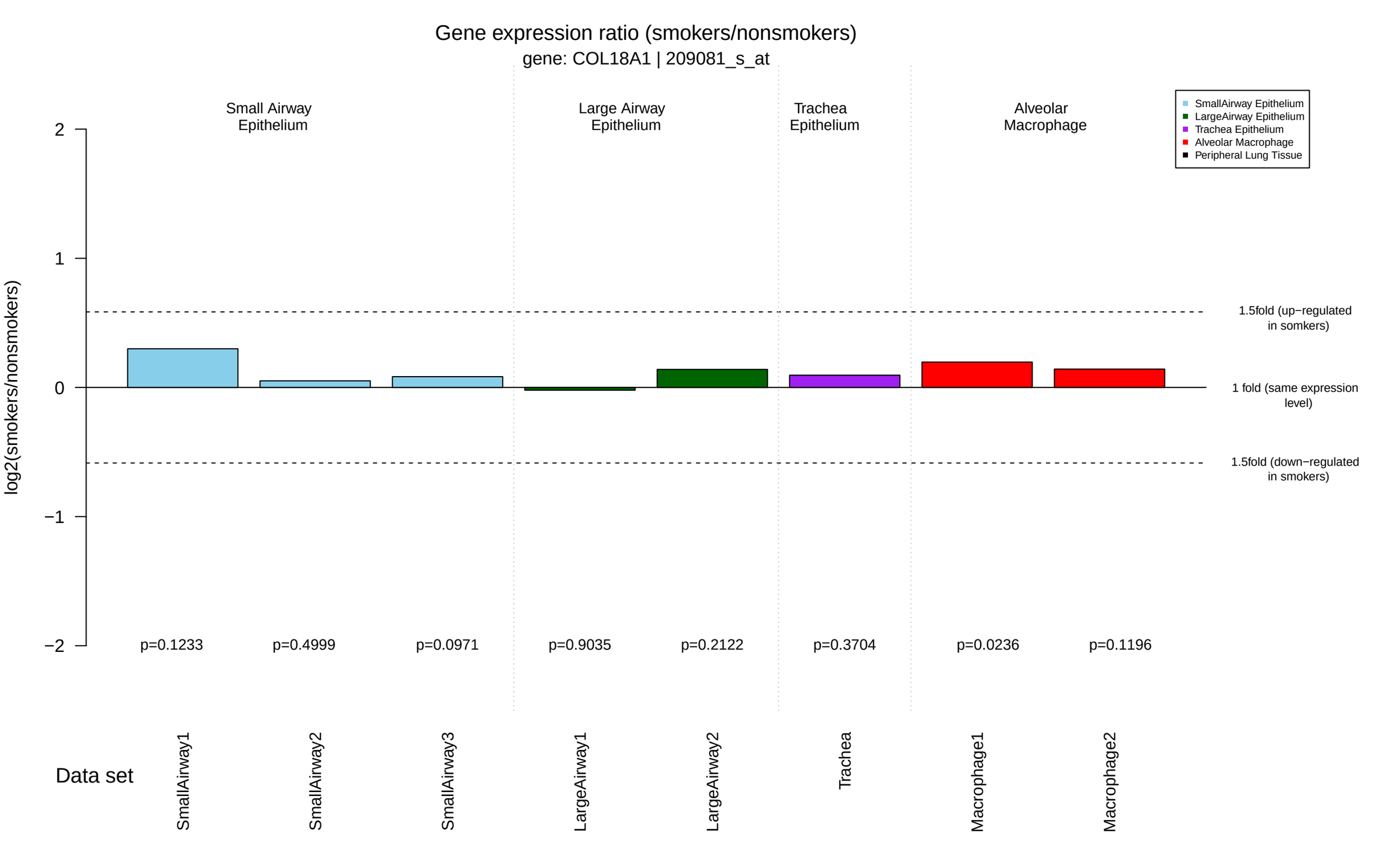
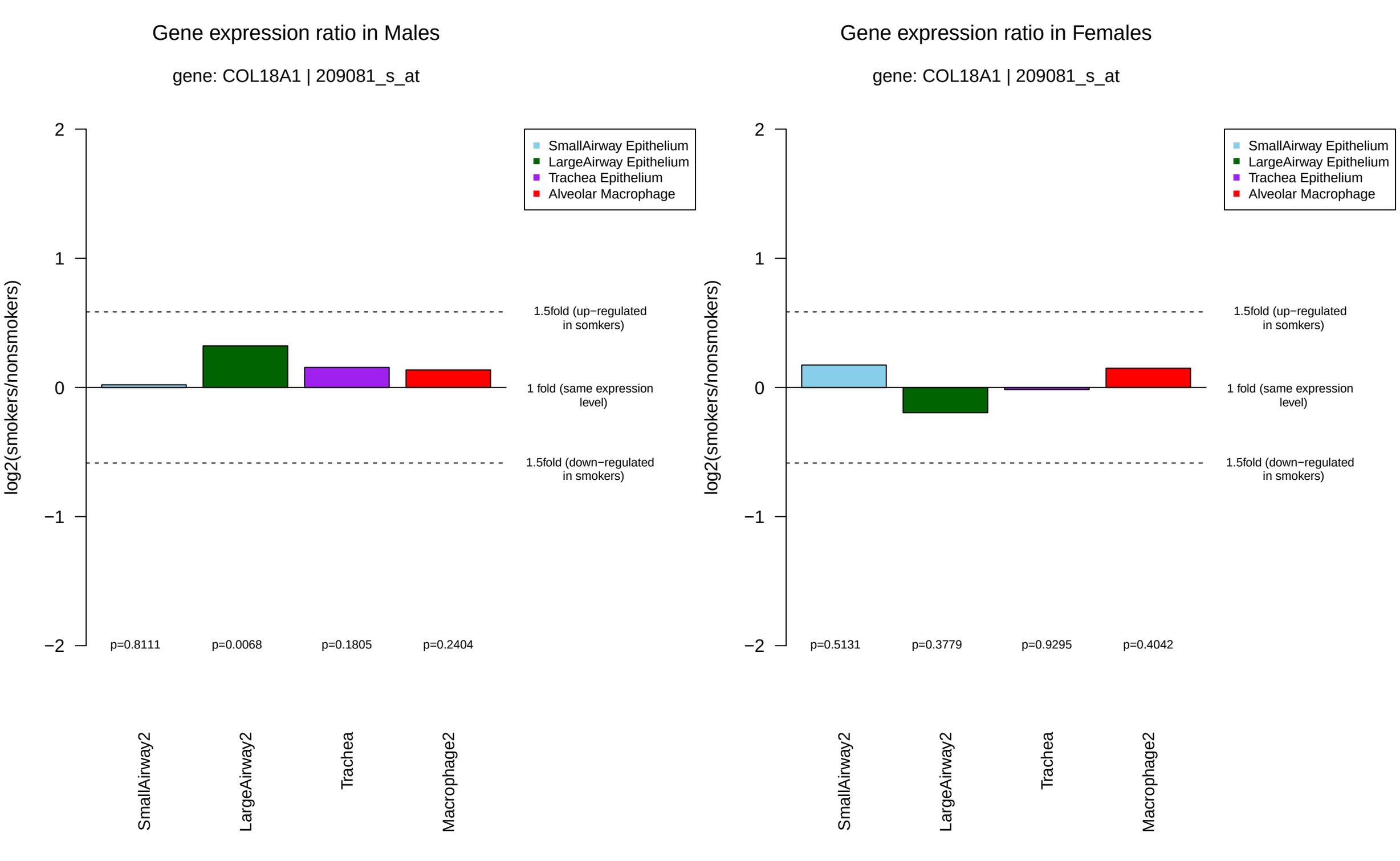
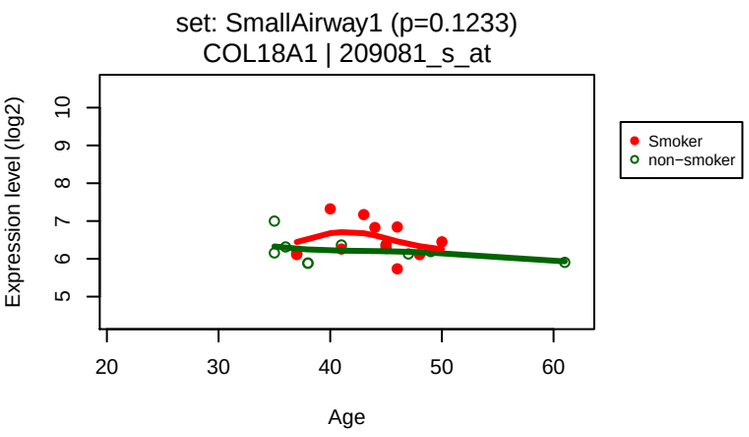
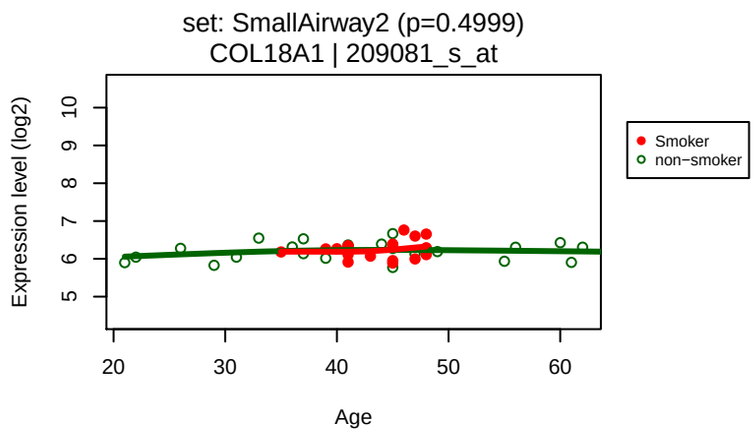
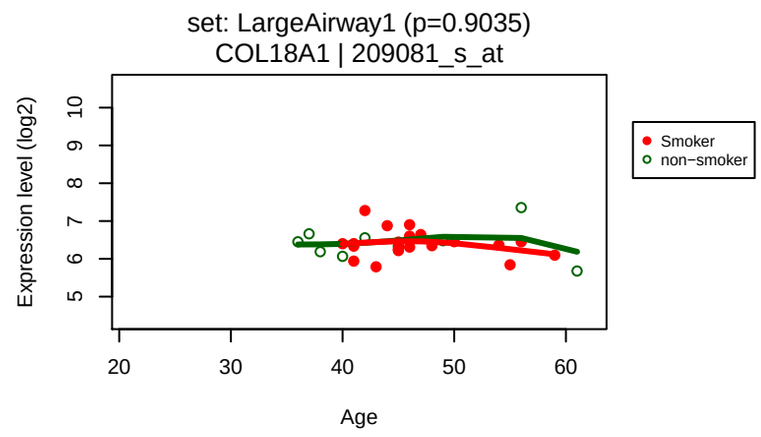

Gene expression of smokers and nonsmokers
gene: COL18A1 | 209081_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1233
p=0.4999
p=0.0971
p=0.9035
p=0.2122
p=0.3704
p=0.0236
p=0.1196
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

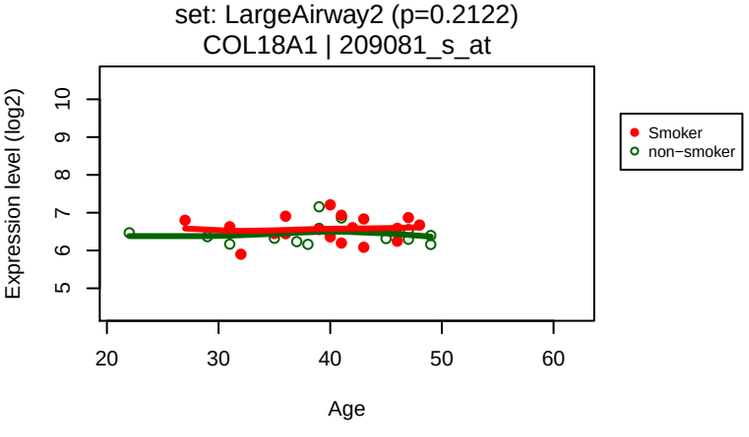

Gene expression ratio (smokers/nonsmokers)
gene: COL18A1 | 209081_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1233
p=0.4999
p=0.0971
p=0.9035
p=0.2122
p=0.3704
p=0.0236
p=0.1196
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

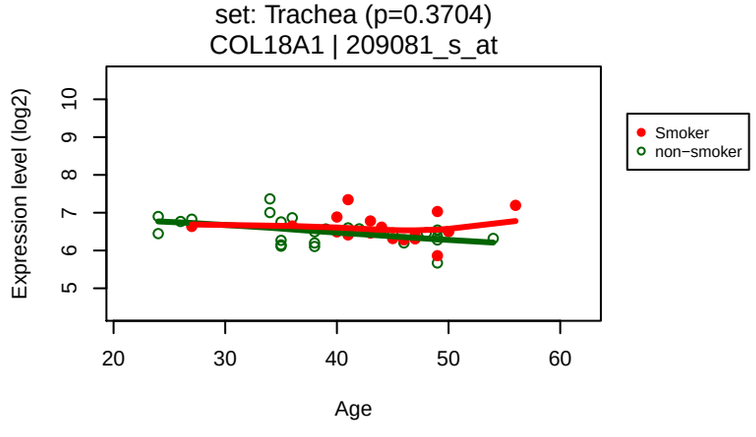

Gene expression ratio in Males
gene: COL18A1 | 209081_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8111
p=0.0068
p=0.1805
p=0.2404
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL18A1 | 209081_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5131
p=0.3779
p=0.9295
p=0.4042
SmallAirway2
LargeAirway2
Trachea
Macrophage2

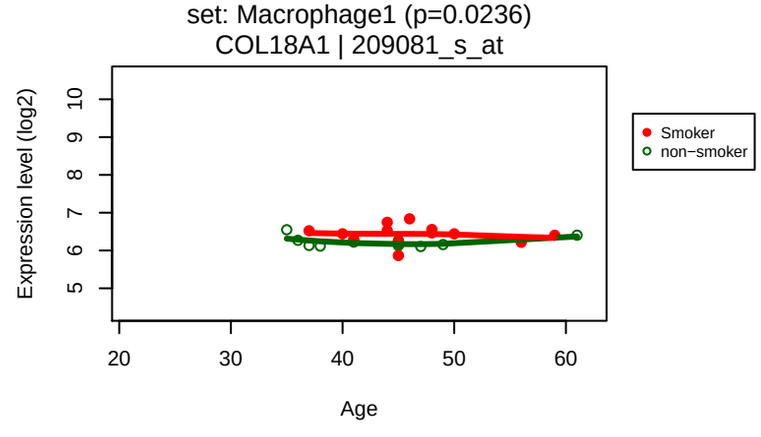

set: SmallAirway1 (p=0.1233)
COL18A1 | 209081_s_at
Expression level (log2)
Age
Smoker
non−smoker

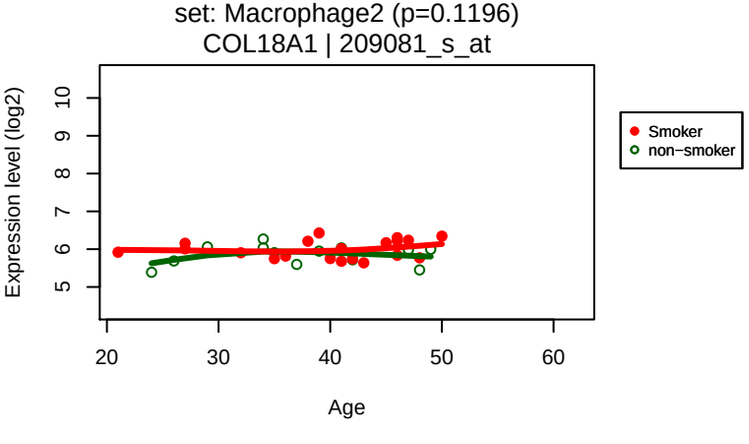

set: SmallAirway2 (p=0.4999)
COL18A1 | 209081_s_at
Expression level (log2)
Age
Smoker
non−smoker

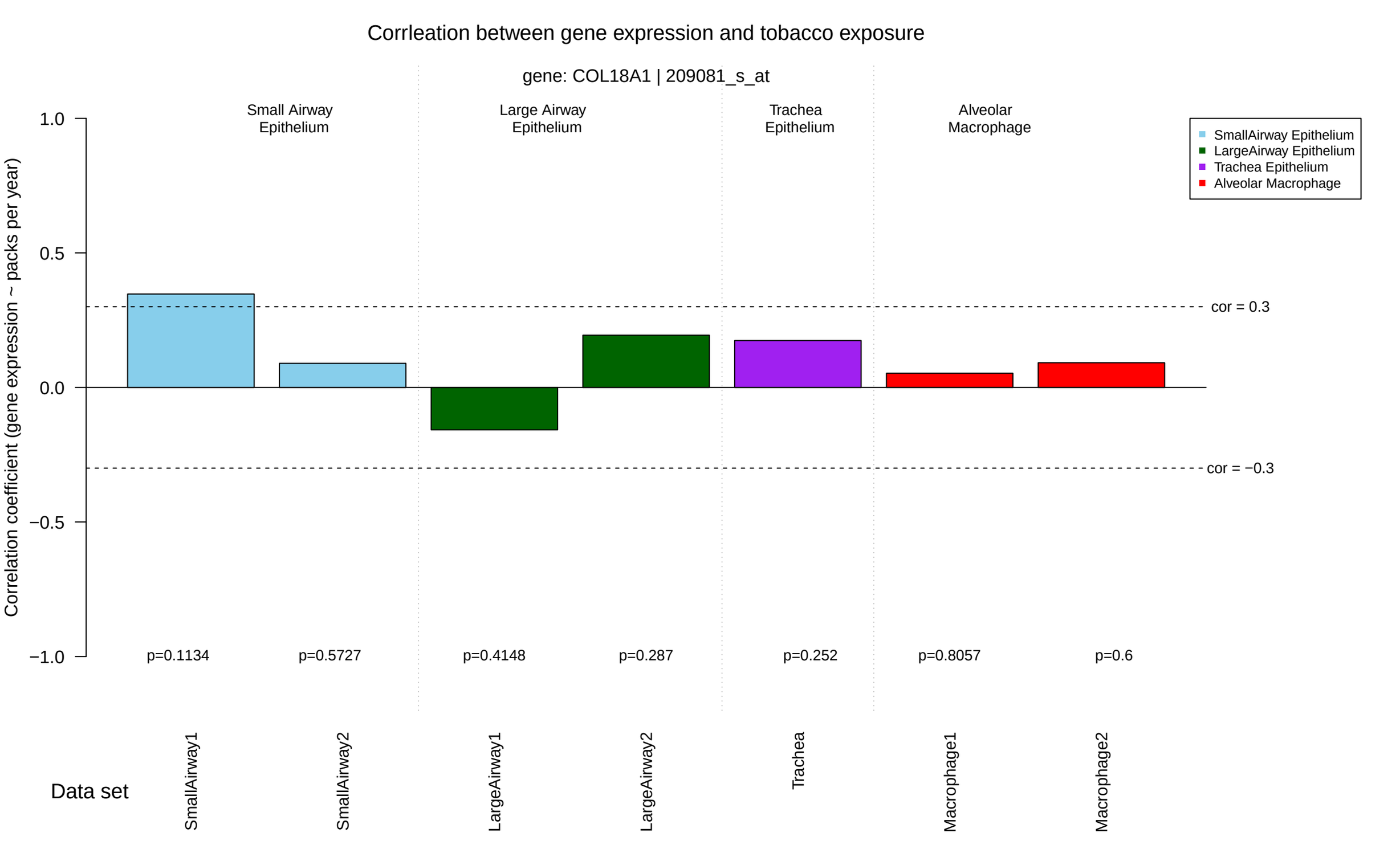

set: LargeAirway1 (p=0.9035)
COL18A1 | 209081_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2122)
COL18A1 | 209081_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3704)
COL18A1 | 209081_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0236)
COL18A1 | 209081_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1196)
COL18A1 | 209081_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL18A1 | 209081_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1134
p=0.5727
p=0.4148
p=0.287
p=0.252
p=0.8057
p=0.6
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2